

Operasi Surds (Penjumlahan & Pengurangan) - Bintang M

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Student

Student: Bintang Nararya P Y

Topic: Operasi Surds (Penjumlahan & Pengurangan)

Status: completed

Lesson Plan

Learning Objectives

- Simplify radical expressions by factoring out perfect squares before performing addition or subtraction.
- Calculate the sum and difference of like surds to solve complex algebraic expressions.
- Apply the order of operations (BODMAS/PEMDAS) to mixed radical expressions involving multiplication and addition.
- Solve multiple-choice problems involving surds with speed and accuracy under time constraints.

Lesson Information

Lesson Sections

1. Spiral Review Warm-up (10 minutes)

Teacher Script:

Hi Bintang! Great to see you. We're jumping straight into the 'Elite Analyst' mode today. Before we hit the new material, let's sharpen your blade with four quick challenges from our previous sessions. Look at the screen. Task 1 & 2: Visualize these surds—can you simplify " 75 In a business context, if a company's revenue is 1.5×10^7 , how do we write that in standard form, and what's the place value of the '5'?

Activities:

- *Visualizing Surds: Simplify " 75 and " 128 .*
- *Business Place Value: Convert 15,000,000 to scientific notation.*
- *Identify the value of the hundred-thousands digit in a \$4,580,000 budget.*

2. Introduction & Real-World Connection (5 minutes)

Teacher Script:

Imagine you are an architect designing a modern skyscraper with triangular glass panels. The sides aren't nice whole numbers; they are surds like $\sqrt{8}$ and $\sqrt{18}$ meters. To do the framing, you can't just guess. You need to know if $\sqrt{8} + \sqrt{18}$ is $\sqrt{26}$ (spoiler: no). Today, you'll learn how to combine these 'irrational' lengths exactly.

Activities:

- Discussion on why we don't use decimals in high-precision engineering/finance.

3. Review of Prior Knowledge (5 minutes)

Teacher Script:

Remember, Bintang: in algebra, you can't add $2x + 3y$. They are different 'families'. Surds work the same way. You can only add $\sqrt{2}$ to another $\sqrt{2}$. But wait—what if one is $\sqrt{8}$? Write down the factors of 8. See a perfect square? Exactly, it's $2\sqrt{2}$. No

Activities:

- Categorizing 'Like Surds' vs 'Unlike Surds'.

4. Problem Introduction (10 minutes)

Teacher Script:

Let's look at a Hard-level problem: $3\sqrt{20} - \sqrt{45} + 2\sqrt{5}$. This looks messy. 1: Simplify every term to its 'prime' surd form. Step 2: Group the like terms. Step 3: Solve. Watch how I break down $3\sqrt{20}$ into $3\sqrt{(4 \cdot 5)}$, which becomes $6\sqrt{5}$. Now, you try simplify

Activities:

- Step-by-step breakdown of a three-term surd expression.
- Application of order of operations (simplification before subtraction).

5. Guided Practice (15 minutes)

Teacher Script:

I'm putting 3 problems on the board. We'll do them 'Rally Style'. I'll do the first step, you do the second. Problem 1: $5\sqrt{12} + 2\sqrt{3}$. Problem 2 (The Distractor): $4\sqrt{18} - 2\sqrt{8} + \sqrt{5}$. Problem 3: A word problem simulation. You have 60 seconds to find the simplified form of $\sqrt{72} + \sqrt{50}$.

Activities:

- Rally Coaching (Alternating steps).
- Timed Challenge: Simplified form of $\sqrt{72} + \sqrt{50}$.

6. Independent Practice (15 minutes)

Teacher Script:

Now, it's your turn to shine as the Lead Consultant. Here are 5 questions. Some have 'traps' where the surds look different but are the same. Focus on speed and precision. I'll be watching your work on the board—go!

Activities:

- *Solve:* $7\sqrt{3} - 2\sqrt{27} + \sqrt{12}$.
- *Solve:* $2\sqrt{50} + 3\sqrt{32}$.
- *Mixed Operation:* $(\sqrt{2} * \sqrt{3}) + \sqrt{54}$ (Note: Multiply first!).

7. Consolidation & Reflection (5 minutes)

Teacher Script:

Excellent work today, Bintang. You've mastered the 'Architecture of Radicals.' Remember: simplify first, then combine like terms. Before you head out, don't forget to log in to your dashboard at <https://learn.algonova.id/login> to access your homework and review today's notes. What was the trickiest part for you today?

Activities:

- *Summary of the 'Simplify-Match-Combine' rule.*
- *Self-reflection on speed.*

Homework

Medium Questions:

1. Simplify: $\sqrt{24} + \sqrt{54}$
2. Simplify: $5\sqrt{2} - \sqrt{18}$
3. Simplify: $3\sqrt{7} + 2\sqrt{28}$
4. Simplify: $\sqrt{20} + \sqrt{45} - \sqrt{5}$

Hard Questions:

1. Simplify: $2\sqrt{48} - 3\sqrt{12} + \sqrt{75}$
2. Simplify: $\sqrt{108} + \sqrt{48} - \sqrt{147}$
3. Simplify: $3\sqrt{2(\sqrt{8} + \sqrt{18})}$
4. Simplify: $(\sqrt{50} - \sqrt{8}) / \sqrt{2}$
5. Find the perimeter of a rectangle with sides $\sqrt{12}$ and $\sqrt{27}$.
6. Calculate: $4\sqrt{20} + 2\sqrt{45} - 3\sqrt{80}$

Answer Key:

$$1. \ 5^6 \ (2^4 = 2^6, \ 5^4 = 3^6)$$

$$2. \ 2^2 \ (5^2 - 3^2)$$

$$3. \ 7^7 \ (3^7 + 4^7)$$

$$4. \ 4^5 \ (2^5 + 3^5 - 1^5)$$

$$5. \ 7^3 \ (8^3 - 6^3 + 5^3)$$

$$6. \ 3^3 \ (6^3 + 4^3 - 7^3)$$

$$7. \ 30 \ (3^2 * 5^2 = 15 * 2)$$

$$8. \ 3 \ (3^2 / 2)$$

$$9. \ 10^3 \ (2 * [2^3 + 3^3])$$

$$10. \ 2^5 \ (8^5 + 6^5 - 12^5)$$

$$\text{REVIEW: Membongkar Surds} - 98 = 7^2$$

$$\text{REVIEW: Membongkar Surds} - 125 = 5^3$$

$$\text{REVIEW: Membongkar Surds} - 200 = 10^3$$

$$\text{REVIEW: Membongkar Surds} - 162 = 9^3$$

$$\text{REVIEW: Nilai Tempat - Standard form of 250 million: } 2.5 \times 10^8$$

$$\text{REVIEW: Nilai Tempat - Value of 7 in \$7,400,000: 7 million}$$

$$\text{REVIEW: Nilai Tempat - Convert 0.00045 to scientific notation: } 4.5 \times 10^{-4}$$

$$\text{REVIEW: Nilai Tempat - If a profit of } 10^6 \text{ is tripled, what is the new value? } 3 \times 10^6$$

Parent Reminder

Bintang's Progress: Radical Operations

Bintang tackled complex radical operations today! We focused on simplifying and combining surds, a key Grade 10 skill. He's learning to apply these to architectural and business contexts. Please ensure he completes the homework on the portal to reinforce these high-level concepts.